

Morning Checklist

Code:

```
from airflow.sdk import dag, task

from airflow import DAG
from airflow.operators.python import PythonOperator, BranchPythonOperator
from airflow.operators.bash import BashOperator
from datetime import datetime, timedelta

def count_tickets():
    import random
    count=random.randint(0, 20)
    print(f"Overnight tickets: {count}")
    return count

def check_workload(**context):
    if context['ti'].xcom_pull(task_ids='count_tickets') >= 10:
        print("High workload detected → Prioritizing urgent tasks")
        return 'busy_day'
    else:
        print("Workload manageable → Proceeding with regular tasks")
        return 'quiet_day'

def busy_day():
    print("BUSY DAY – calling in extra support staff.")
def quiet_day():
    print("QUIET DAY – normal staffing today.")
def daily_summary():
    print("Morning checklist complete. Have a good day!")

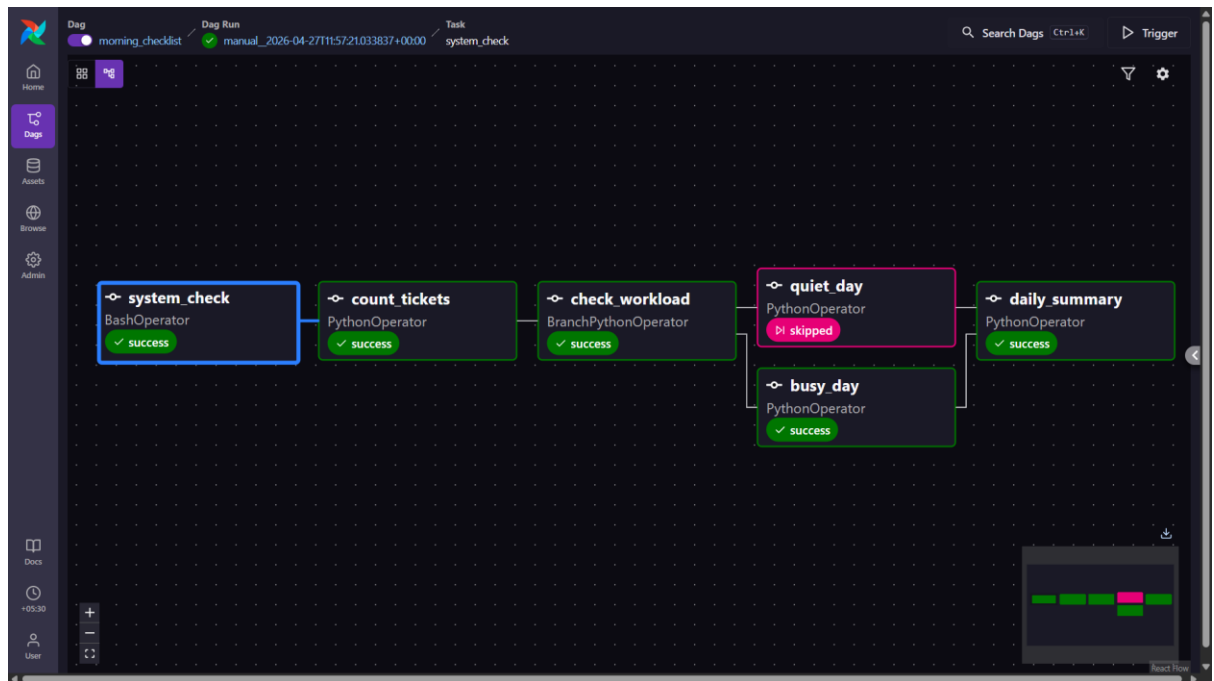
with DAG(
    dag_id="morning_checklist",
    schedule="@daily",
    start_date=datetime(2026, 1, 1),
    catchup=False
)as dag:

    system_check =BashOperator(
        task_id="system_check",
        bash_command="echo 'Office system check – all OK' && date"
    )

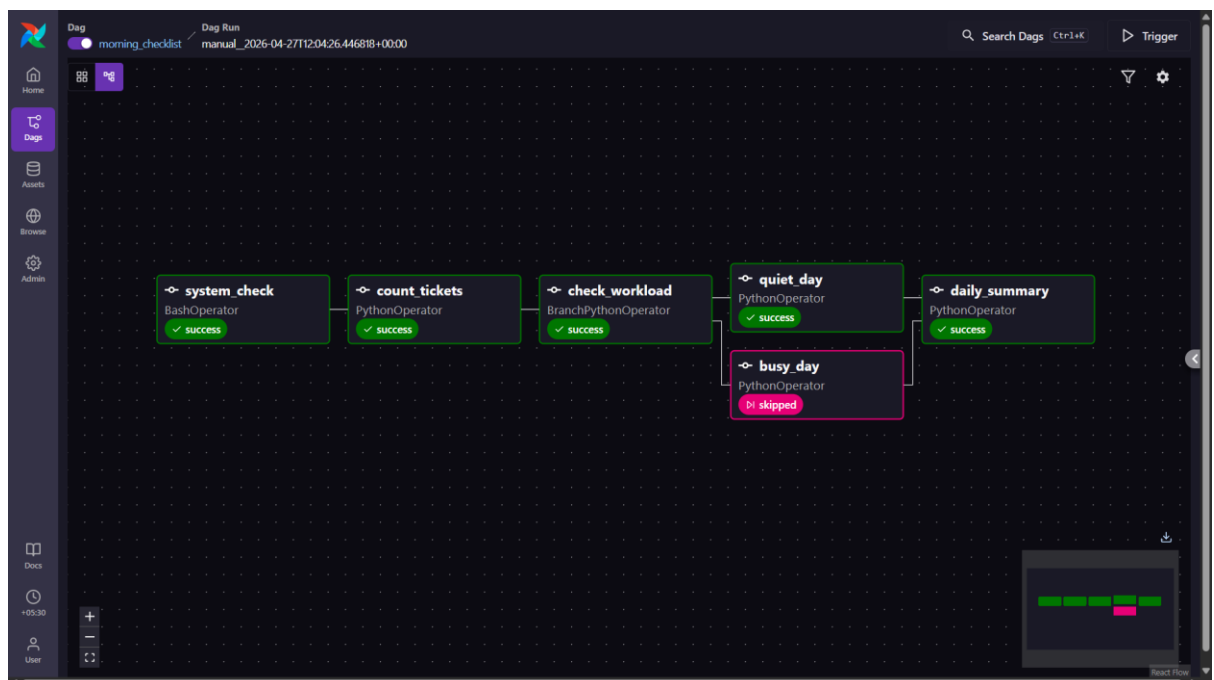
    count_tickets=PythonOperator(
        task_id="count_tickets",
        python_callable=count_tickets
    )
```

```
check_workload=BranchPythonOperator(  
    task_id="check_workload",  
    python_callable=check_workload  
    # provide_context=True  
)  
  
busy_day=PythonOperator(  
    task_id="busy_day",  
    python_callable=busy_day  
)  
  
quiet_day=PythonOperator(  
    task_id="quiet_day",  
    python_callable=quiet_day  
)  
  
daily_summary=PythonOperator(  
    task_id="daily_summary",  
    python_callable=daily_summary,  
    trigger_rule="none_failed"  
)  
  
system_check>>count_tickets>>check_workload>>[busy_day,quiet_day]>>daily_s  
ummary
```

1. Graph View — busy_day is GREEN, quiet_day is pink (skipped), daily_summary is GREEN



2. Graph View — quiet_day is GREEN, busy_day is pink (skipped), daily_summary is GREEN



3. Task Log of system_check — showing "Office system check — all OK" and the date from the bash command

The screenshot displays the Apache Airflow web interface. On the left, a sidebar contains navigation links for Home, Dags, Assets, Browse, Admin, Docs, and a clock showing +05:30. The main area shows a DAG named 'morning_checklist' with a 'Dag Run' status of 'manual_2026-04-27T12:04:26.446818+00:00'. The task 'system_check' is highlighted, showing a 'Success' status. The task details panel on the right shows the operator as 'BashOperator', start and end dates as '2026-04-27 17:34:26', duration as '00:00:00.137', and DAG version as 'v1'. The 'Logs' tab is active, displaying a list of log messages. A red box highlights the following log entries:

```
ls sources=["/usr/local/airflow/logs/dag_id-morning_checklist/run_id-manual_2026-04-27T12:04:26.446818+00:00/task_id-system_
6] WARNING - Astro managed secrets backend is disabled
6] INFO - DAG bundles loaded: dags-folder
6] INFO - Filling up the DagBag from /usr/local/airflow/dags/Assignment-2/checklist.py
6] WARNING - The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'airflow.providers.standard.
6] WARNING - The 'airflow.operators.python.BranchPythonOperator' attribute is deprecated. Please use 'airflow.providers.stai
6] WARNING - The 'airflow.operators.bash.BashOperator' attribute is deprecated. Please use 'airflow.providers.standard.oper
6] INFO - TaskInstance Details: dag_id=morning_checklist task_id=system_check dagrun_id=manual_2026-04-27T12:04:26.446818+00:00
6] INFO - Temp dir root location: /tmp
6] INFO - Running command: ['/usr/bin/bash', '-c', "echo 'Office system check - all OK' && date"]
6] INFO - Output:
6] INFO - Office system check - all OK
6] INFO - Mon Apr 27 12:04:26 UTC 2026
6] INFO - Command exited with return code 0
6] INFO - Pushing xcom ti=RuntimeTaskInstance(id=UUID('019dced3-96dc-7df7-97c2-a37c3b0973dd'), task_id='system_check', dag_id='morning_checklist')
```

4. Task Log of check_workload — showing the ticket number and the branch decision printed

The screenshot displays the Apache Airflow web interface. On the left, a sidebar contains navigation links for Home, Dags, Assets, Browse, Admin, Docs, and a clock showing +05:30. The main area shows a DAG named 'morning_checklist' with a 'Dag Run' status of 'manual_2026-04-27T12:09:02.717273+00:00'. The task 'count_tickets' is highlighted, showing a 'Success' status. The task details panel on the right shows the operator as 'PythonOperator', start and end dates as '2026-04-27 17:39:04', duration as '00:00:00.113', and DAG version as 'v1'. The 'Logs' tab is active, displaying a list of log messages. A red box highlights the following log entries:

```
> Log message source details sources=["/usr/local/airflow/logs/dag_id-morning_checklist/run_id-manual_2026-04-27T12:09:02.7
2 [2026-04-27 17:39:04] WARNING - Astro managed secrets backend is disabled
3 [2026-04-27 17:39:04] INFO - DAG bundles loaded: dags-folder
4 [2026-04-27 17:39:04] INFO - Filling up the DagBag from /usr/local/airflow/dags/Assignment-2/checklist.py
5 [2026-04-27 17:39:04] WARNING - The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'air
6 [2026-04-27 17:39:04] WARNING - The 'airflow.operators.python.BranchPythonOperator' attribute is deprecated. Please use '
7 [2026-04-27 17:39:04] WARNING - The 'airflow.operators.bash.BashOperator' attribute is deprecated. Please use 'airf
8 [2026-04-27 17:39:04] INFO - TaskInstance Details: dag_id=morning_checklist task_id=count_tickets dagrun_id=manual_
9 [2026-04-27 17:39:04] INFO - Overnight tickets: 7
10 [2026-04-27 17:39:04] INFO - Done. Returned value was: 7
11 [2026-04-27 17:39:04] INFO - Pushing xcom ti=RuntimeTaskInstance(id=UUID('019dced7-ce0d-74e9-8f98-fa476dea239d'), ta
```

Home

Dags

Assets

Browse

Admin

Docs

+05:30

User

Dag

morning_checklist

Dag Run

manual_2026-04-27T12:09:02.717273+00:00

Task

check_workload

Search Dags

Ctrl+K

Trigger

check_workload

Success

Operator

BranchPythonOperator

Start Date

2026-04-27 17:39:05

End Date

2026-04-27 17:39:05

Duration

00:00:00.140

Dag Version

v1

Logs

Rendered Templates

XCom

Asset Events

Audit Log

Code

Details

All Log Levels

All Sources

Log message source details sources=["/usr/local/airflow/logs/dag_id-morning_checklist/run_id-manual_2026-04-27T12:09:02.717273+00:00"]

2 [2026-04-27 17:39:05] WARNING - Astro managed secrets backend is disabled

3 [2026-04-27 17:39:05] INFO - DAG bundles loaded: dags-folder

4 [2026-04-27 17:39:05] INFO - Filling up the DagBag from /usr/local/airflow/dags/Assignment-2/checklist.py

5 [2026-04-27 17:39:05] WARNING - The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use ''

6 [2026-04-27 17:39:05] WARNING - The 'airflow.operators.python.BranchPythonOperator' attribute is deprecated. Please use ''

7 [2026-04-27 17:39:05] WARNING - The 'airflow.operators.bash.BashOperator' attribute is deprecated. Please use ''

8 [2026-04-27 17:39:05] INFO - TaskInstance Details: dag_id=morning_checklist task_id=check_workload dagrun_id=manual_2026-04-27T12:09:02.717273+00:00

9 [2026-04-27 17:39:05] INFO - Workload manageable -> Proceeding with regular tasks

10 [2026-04-27 17:39:05] INFO - Done. Returned value was: quiet_day

11 [2026-04-27 17:39:05] INFO - Branch into quiet_day

12 [2026-04-27 17:39:05] INFO - Following branch ('quiet_day')

13 [2026-04-27 17:39:05] INFO - Skipping tasks [('busy_day', -1)]

14 [2026-04-27 17:39:05] INFO - Skipping downstream tasks.

heck

count_tickets

check_workload

PythonOperator

BranchPythonOperator

Pyt

React Flow